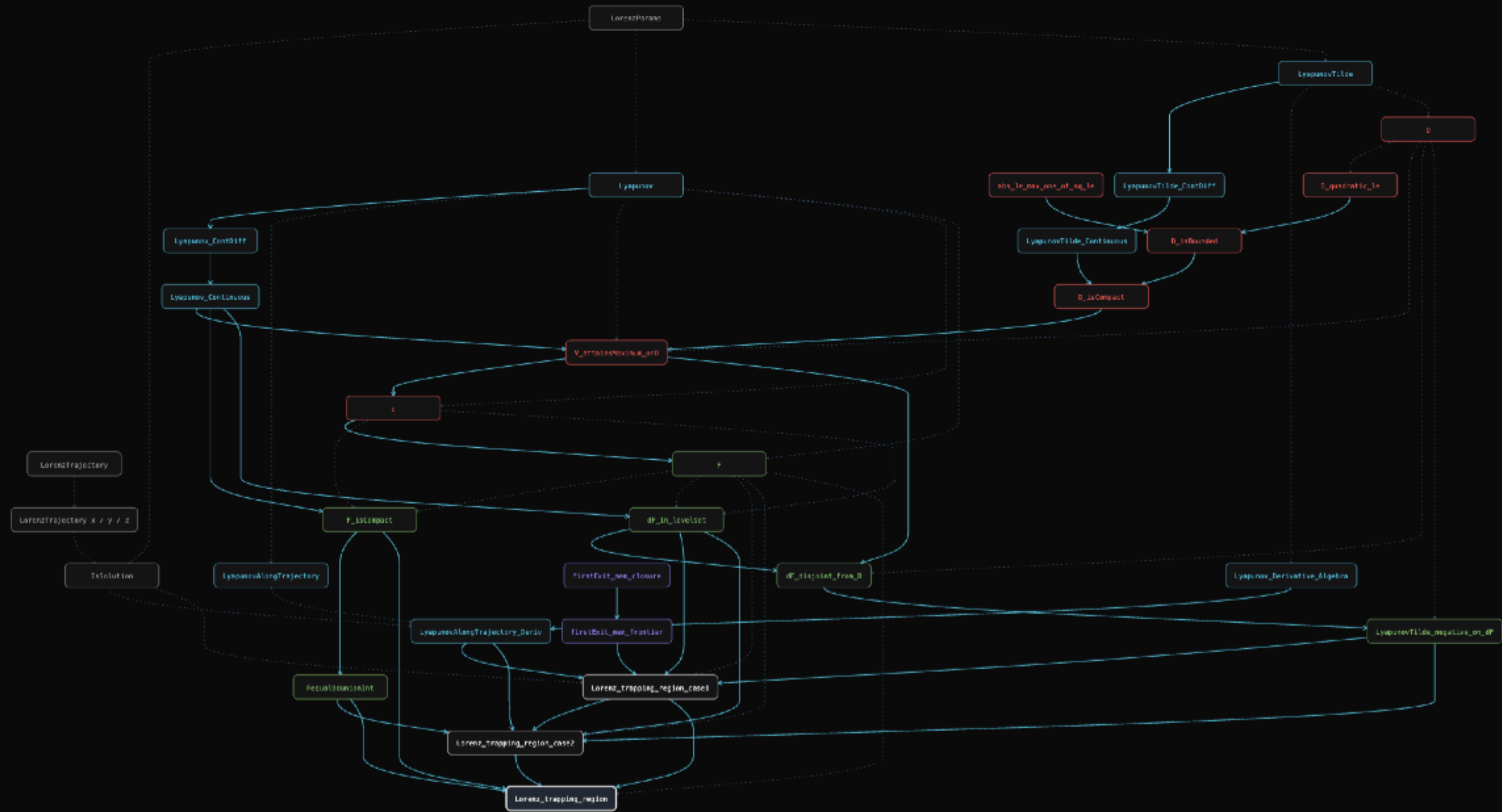


- Phase 1 · The dynamical system
- Phase 2 · Defining V and $\tilde{V}$
- Phase 3 · The set D and $\sup c$
- Phase 4 · The set E and ∂E
- Toolkit · replaces Case 1 subcases 1–2
- Phase 5 · Trapping region



STEP 31 OF 31 · THEOREM · PHASE 5 · THEOREM

Lorenz_trapping_region

"If $\phi(t_0) \in E$, then $\phi(t) \in \text{int}(E)$ for all $t > t_0$." Four moves: use $F\ p$ to supply the witness, $F_isCompact.isBounded$ to discharge the boundedness conjunct, $F_equalBdunionInt$ to split F into interior u frontier, and then each piece to its case. Note that the conclusion is **stronger than the hypothesis**: land anywhere in F — even exactly on the boundary — and at every later instant you are strictly inside. The trapping region does not merely hold you; it pulls you off its own edge.

```
theorem Lorenz_trapping_region (p) (φ)
  (hφ : IsSolution p φ) :
  ∃ F : Set (ℝ × ℝ × ℝ),
    Bornology.IsBounded F ∧
    ∀ t0 : ℝ, φ.total t0 ∈ F →
      (∀ t > t0, φ.total t ∈ interior F) := by
    use F p
    rw [and_iff_left ?_]
    · refine IsCompact.isBounded ?_
      apply F_isCompact
    · intro t0 h1
```


Next Steps

① Contribute the boundary lemmas to Mathlib.

Goal. Generalize and upstream the auxiliary topological results developed for the trapping region proof.

- `firstExit_mem_frontier` *The first exit time lies on the frontier of the set.*
- `firstExit_mem_closure` *A right-continuous flow maps the first exit time into the closure of the complement.*

These lemmas are completely independent of the Lorenz system and are potentially useful in broader formalizations.

② Extend the formalization of the Lorenz system.

Prove additional fundamental results, including

- Global Existence Theorem
- Existence of an Attracting Set
- Further qualitative properties of the Lorenz flow.

③ Generalize the trapping region lemma.

Develop a version applicable to more general dynamical systems, including

- Higher-dimensional systems
- Non-smooth flows
- Infinite-dimensional systems

⇒ Towards a reusable formal library for trapping regions and dynamical systems in Lean.